The Influence of Heterogeneous Skier Decisions on Lift-Line Congestion and Queue Stability

Adler J. Linfoot

Courant Institute

Abstract and Context

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Brief Synopsis

Common Questions, the Answers to Assignment 1

Abstract and Context

Every skier plays some version of the same game: “How do I maximize my enjoyment of the mountain?” For many, the answer is straightforward: maximize the time actually spent skiing.

In recent years, ski resorts have faced increasing criticism over long lift lines, unreliable operations, and the perceived inadequacy of the experience relative to rising ticket prices. Yet lift-line congestion is not simply a function of the number of skiers on a mountain or the capacity of its infrastructure. It is also influenced by how skiers distribute themselves across the resort.

Consider a skier arriving at the bottom of a run and encountering a long lift line. They face a simple decision: enter the queue immediately or travel to another lift in hopes of finding a shorter wait. The alternative, however, carries uncertainty. While the skier can observe the queue directly in front of them, their knowledge of other lift lines may be incomplete or outdated.

Apply that behavioral pattern to hundreds of skiers independently, each making similar decisions attempting to maximize their own time on the mountain. Their collective behavior may produce congestion patterns that no individual skier intended.

This project investigates how heterogeneous skier decision-making influences lift-line congestion and queue stability. Using a simplified system of population-level differential equations and an agent-based simulation of a ski resort, it examines how individual terrain preferences, remembered queue lengths, and access to alternative lifts affect the distribution of skiers across the mountain.

Note: This paper uses the term “skier” as an analog for both skiers and snowboarders in addition to others who use the mountain such as monoskiers, and snowfeet

Methodology

The model is constructed progressively, beginning with a generalized mathematical representation of an individual skier and concluding with a population-level description of lift-line congestion; first defining the skier's objective, introducing empirically motivated heterogeneity, then establishing class-specific behavioral constraints. After which, movement through the resort and individual decision-making under incomplete information are both formalized.

For the initial descriptions, lift waiting times are assumed constant. This assumption is subsequently relaxed to account for the interaction of individual decisions, allowing queue populations to evolve dynamically. The resulting equations provide the foundation for a stochastic agent-based simulation.

2.1 General Skier Model

Consider an individual skier *i* participating in a finite operating day of duration *H*. Assume that the skier seeks to maximize the utility derived from time spent skiing, subject to time consumed by lift queues, uphill transportation, and transit between lifts. This can be expressed as such:

$$T_{skiing} = H - \left(\sum_{x=0}^{y} T_{queue} + \sum_{x=0}^{y} T_{uphill} + \sum_{j=0}^{l} T_{transit}\right)$$

Moreover, the skier's available time is constrained by the amount of time they spend on the resort, modifying *H*. Henceforth *H*, when unmodified, can be interpreted as the upper bound of time that an individual has to ski the resort. The general skier model makes no assumptions

regarding terrain preferences, ability, or willingness to change routes. These characteristics are introduced through class-specific restrictions.

2.2 Empirical Motivation for Skier Heterogeneity

A homogeneous population is insufficient to represent differences in terrain accessibility, route preference, and willingness to relocate.

Schlemmer and Peters published a research paper depicting several unique skier phenotypes. These observations motivate the introduction of heterogeneous skier populations. The present model distinguishes four behavioral classes: all-mountain skiers, vacationers, lappers, and mixed skiers derived from said article[1].

For the purposes of this paper, the proportion of the resort population belonging to classes is as follows:

All-Mountain Skiers	25%
Vacationers	35%
Lappers	15%
Mixed Skiers	25%

Note that these population percentages are assumed, not empirically estimated. The class proportions are treated as model parameters, with their values stated explicitly and varied where appropriate.

2.3 Class-Specific Skier Models

The generalized objective in Section 2.1 is modified through class-dependent terrain preferences and admissible route choices.

All-Mountain Skiers	May access all modeled terrain and change lifts in response to perceived waiting-time advantages.
Vacationers	Are restricted to terrain compatible with their skiing ability. Their access to an additional lift depends on the resort's operating configuration.
Lappers	Prefer a particular run or group of runs and remain committed to their associated lift, irrespective of congestion elsewhere.
Mixed	Initially explore the mountain before developing preferences for particular runs. They retain the ability to relocate when waiting-time advantages justify doing so.

For the purposes of the variable assignment, let $A_i(t)$ denote the set of lifts accessible to skier i.

2.4 Mountain Topology and Temporal Constraints

The resort is represented as a network consisting of twenty ski runs and four lifts, with three or four lifts operational according to the experimental configuration. Each skier traverses the network through a sequence of downhill runs, lift queues, uphill transportation, and inter-lift transfers with the duration of a traversal depending on the distance traveled and the mode of travel. The speed of each skier, with respect to mode, is held uniform for the simplicity of modeling.

Note: A reasonable addition to this simplification would be a dynamic movement system that scales skier speeds at the skill level or individually at the agent level.

2.5 Skier Decision-Making Under Incomplete Information

Upon entering the resort (spawning), each skier observes the waiting times of all operational lifts and stores these observations in memory.

Thereafter, the skier directly observes only the queue at their current location. The waiting times at other lifts constitute hidden states, represented by their most recently observed values.

Each individual skier *i* possesses its own kernel containing the last-observed queue size at every lift which serves terrain the skier may ski. Within this kernel, the skier *i* chooses to migrate and enter the line in which the sum of the travel time, from their current location, to that lift plus the last remembered queue time is the least.

$$j^{*}(t) = \underset{j \in A_i(t)}{arg\,min} (T_{travel\, c \to j} + k_j)$$

Observe the equation denotes the optimization function *arg min* as the summation of the travel time from lifts *c*, *c* being the arbitrary denotation of the lift the agent currently is at, to lift *j* plus the memory vector of lift line queue time at that lift *j* denoted by k_j. If multiple lifts share an equal estimated cost, the skier randomly chooses among tied admissible lifts.

Note a remembered observation is updated only when the skier revisits the corresponding lift. Consequently, the accuracy of stored information may deteriorate as the resort's actual queue conditions change. This information staleness is a deliberate feature of the model. It permits decisions that are rational relative to the skier's available information to be suboptimal relative to the true state of the mountain.

Once a destination has been selected, the skier commits to that decision until reaching the destination. Decisions are reconsidered only at subsequent eligible decision points occurring after a descent of their chosen path has been completed.

2.6 Dynamic Lift-line Congestion

Queues are populated by the same skiers making the decisions described above changing the size and time required for a skier to board, ascend, and ultimately ski.

A skier's decision to remain at a lift contributes to its queue, while a decision to relocate changes the distribution of future arrivals across the resort. These changes influence subsequent waiting times and therefore the information upon which future decisions are based.

Now the constant-queue assumption is relaxed and treats each lift line as a dynamic population. Let $Q_j(t)$ denote the number of skiers waiting at lift j, $\lambda_j(t)$ its arrival rate, and $S_j(t)$ its boarding rate for that lift. Henceforth the population dynamics are governed by:

$$\frac{dQ_j}{dt} = \lambda_j(t) - S_j(t)$$

Where lift service is constrained by its maximum boarding capacity and the quantity of waiting skiers. Note that $0 \leq S_j(t) \leq \mu_j$, $S_j(t), \lambda_j(t) \geq 0$, and that the quantity of skiers cannot be negative.

Queue size, Q_j, dictates queue times, k, the time for an individual to arrive at the lift line before boarding, as a linear relationship with constant μ_j representing the fixed rate at which the ski lift throughput at lift j. Note that $S_j(t)$ is not used, as a queue may be empty creating a divide by zero error.

$$k_j(t) = \frac{Q_j}{\mu_j}$$

Realized arrivals are independently decided by each agent, henceforth a uniform arrival formula is generally inappropriate. Note a simplified aggregate closure is required for analytical stability calculations.

2.7 Agent-Based Implementation and Experimental Design

The mathematical formulation is implemented as a stochastic agent-based simulation in which individual skier movements and decisions generate the observed aggregate arrival and departure rates. The interaction between skier preferences, incomplete information, and lift capacity produces the population-level queue dynamics described in Section 2.6.

The aggregate system is examined for equilibrium conditions, defined by the equality of arrival and departure rates at each lift. The stability of these equilibria is investigated under perturbations to queue populations, using a simplified aggregate representation of skier arrivals where necessary. Changes in equilibrium behavior and stability across operating configurations are subsequently examined.

The principal experimental intervention is the activation of a fourth lift, changing both the resort's total transportation capacity and the set of routes accessible to eligible skiers. The three-lift and four-lift configurations are compared to examine their effects on congestion and queue stability. Additional comparisons distinguish the effects of increased lift capacity from those of expanded skier accessibility.

Further experiments vary skier arrival rates, population composition, and terrain-access restrictions while maintaining otherwise comparable operating conditions.

The primary observables are individual lift queue lengths, estimated and realized waiting times, queue variability, skier relocation behavior, and cumulative time spent skiing. These measurements allow the effects of individual decision-making to be evaluated at both the skier and resort-wide population levels.

A minimum of twenty stochastic replications is conducted for each parameter configuration. Comparisons across repeated simulations are used to distinguish systematic changes in queue behavior from stochastic variation.

Finally, the resulting queue trajectories are compared with the deterministic population-level formulation to assess the extent to which the aggregate mathematical model captures the dynamics generated by heterogeneous skier decisions under incomplete information.

References

Schlemmer, Patrick, and Katharina Peters. "Rethinking Growth Orientation: Do Ski-Resorts' Strategies Fit the Profiles of Today's Alpine Skiers?" Journal of Tourism Research, 2024. opus4.kobv.de/opus4-hm/frontdoor/deliver/index/docId/559/file/JTR_JTR2708.pdf. Accessed 26 Oct. 2026.

Alexis Poulhes, Paul Mirial. "Dynaski, an Agent-based Model to Simulate Skiers in a Ski Area." Procedia Computer Science, vol. 108, 2017, pp. 2334–2343. ScienceDirect, https://www.sciencedirect.com/science/article/pii/S1877050917309559. Accessed 27 Oct. 2026.